

EXAM #1 INFORMATION

Where and when: Wednesday October 15, 5:45-7:15 PM, Van Vleck B130.

Structure of exam: There could be true/false questions and there could be some short answer questions (e.g. what is the max/min/sup/inf of a given set). The rest of the exam will be proof questions similar to those on the homework sets and the activities.

How to prepare: Redoing the activities and homework problems is a good start. It is also a good idea to read the posted solutions for the homework problems. Two old exams are posted to the canvas page for further practice. Note that the structure & point values will be similar to, but potentially not exactly the same as, the old exams. After that, the textbook has additional problems in each section that you could use as practice (selected hints and answers are given in the backs).

Material tested: The exam will cover “Chapter 0”, Chapter 1 in Ross, and part of Chapter 2 in Ross (up to and including Example 3). Sections 5 and 6 in Chapter 1 were not discussed in class and are not on the exam.

The important Concepts & Learning Objectives for each chapter are given below.

Chapter 0:

- Determine the truth value of a given statement.
- Provide counterexamples for statements that are false.
- Negate specific statements.
- Use a direct proof to prove that a given statement is true.
- Identify logical fallacies or gaps in logic in incorrect proofs.
- Understand and be able to write proofs by cases, contradiction, and contrapositive.
- Identify what proof technique is most appropriate depending on the proposition to be proven.
- State the definition of a set and the empty set.
- Prove one set is a subset of another.
- Prove two sets are equal to each other.
- Understand the set operations of intersection, union, complement, and set minus.

Chapter 1:

Date: October 7, 2025.

- Understand and be able to write proofs by induction. This includes identifying the base case and inductive step required in proofs by induction.
- Distinguish and understand the relationships between the various sets of numbers, natural numbers, integers, rational numbers, irrational numbers, and the real numbers.
- Prove that a number is irrational using a proof by contradiction.
- Prove properties of the absolute value function using cases.
- State and use the definitions of the supremum and infimum of a set.
- Identify the maximum, minimum, supremum, and infimums of a variety of sets or state they do not exist.
- Know and be able to use the Completeness Axiom, Archimedean Property, and the Density of the rational numbers.

Chapter 2:

- State the formal definition of a sequence.
- State the precise definition of the limit of a sequence and know what it means for the definition not to hold.
- Give a formal proof that an explicit sequence, e.g. $(\frac{1}{n})$ or $((-1)^n)$, converges to a specific number or diverges.
- Give a formal proof that an abstract sequences converges (e.g. prove: if $\lim s_n = 0$ and $s_n \geq 0$ for all n , then $\lim \sqrt{s_n} = 0$).